

CONSTRUCTION OS

PRD - Mobile & Offline Platform Architecture

Implementation-ready Product Requirements Document for Product, UX/UI, Mobile, Frontend, Backend, Platform, Security, QA, DevOps and Field Operations Teams

Product purpose: Provide a secure, permission-aware field execution environment that caches authorized project data, captures evidence and business activity offline, and synchronizes it through validated, idempotent commands without creating parallel sources of truth.

Document control	Value
Document	PRD_Mobile_Offline_Platform
Version	1.0
Status	Implementation-ready functional and technical specification
Audience	Product, UX/UI, mobile, frontend, backend, platform, security, QA, DevOps, SRE and field operations teams
Primary surfaces	Responsive web, PWA, native mobile, tablet, portal mobile experiences
Core architecture	Permission-safe mobile projections + encrypted offline workspaces + command-based synchronization + conflict-safe server validation
Access model	Deny-by-default, tenant/company/project/scope/device aware, effective-dated, revalidated on sync and fully auditable
Source-of-truth policy	Operational modules own records and business rules; Mobile & Offline owns device delivery, caching, local drafts, synchronization and conflict handling
Initial integration policy	Self-contained initial release. External MDM, maps, biometric hardware, storage and messaging integrations remain future roadmap items
Platform Configuration	Mandatory for module availability, workspace scope, expiry, storage, GPS, file capture, sync, security, conflicts, navigation and device support
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1. Executive Summary

The Mobile & Offline Platform is the field-operating layer of ConstructionOS. It enables authorized users to view controlled project information, complete forms, record progress, inspect work, report safety events, capture evidence and prepare transactions when connectivity is weak, intermittent or unavailable. It is a platform-wide capability rather than an independent business module.

The design uses permission-safe mobile projections, encrypted offline workspaces and command-based synchronization. Offline users never edit authoritative database rows directly. They create validated commands carrying expected server versions, device identity, timestamps, evidence and idempotency keys. The server rechecks permissions and current business rules before applying each command. Critical conflicts are never resolved silently.

Permanent rule: Operational modules own business truth. Mobile & Offline owns device delivery, local caching, draft capture, secure storage, synchronization, conflict detection and controlled revocation.

1.1 Required outcomes

- Keep essential field workflows usable during weak or unavailable connectivity.
- Prevent duplicate records, double submissions and unsafe last-write-wins behavior.
- Protect company, project, employee, client, supplier and document data on mobile devices.
- Provide transparent states for local-only, pending, synchronized, rejected and conflicted records.
- Allow administrators to revoke devices, workspaces and permissions without deleting audit history.
- Use optimized mobile projections instead of unrestricted operational database access.
- Support mobile web immediately and advanced native offline operations through a shared architecture.

1.2 Primary implementation decision

Architecture decision: Treat every offline mutation as a durable command that is validated and applied by the owning source module. Never synchronize by blindly copying local database state to the server.

2. Product Vision and Business Outcomes

2.1 Product vision

ConstructionOS should remain operational where construction work actually happens: in basements, remote sites, unfinished towers, warehouses and moving delivery areas. The field user must be able to open assigned work quickly, capture reliable evidence, understand what is still local and trust that the system will synchronize safely when connectivity returns.

2.2 Business outcomes

- Reduce paper forms, later transcription and uncontrolled messaging-app evidence.
- Improve timeliness and completeness of progress, quality, HSE, asset, inventory and labour records.
- Lower rework caused by outdated drawings, missing context and lost field notes.
- Provide stronger evidence chains through device, timestamp, location and file checksums.
- Reduce field downtime caused by temporary network outages.
- Create a consistent mobile foundation reused by every ConstructionOS module and portal.
- Allow secure scaling across companies, projects, devices and countries without independent mobile databases.

2.3 Failure risks to prevent

Risk	Required prevention
Offline copy becomes a second source of truth	Use projections and commands; server remains authoritative.
Same action is uploaded repeatedly	Mandatory idempotency keys and deterministic server responses.

Risk	Required prevention
User believes emergency report was delivered	Distinct local-recorded and server-received states with explicit warning.
Revoked data remains searchable	Local access grants, expiry checks, index removal and encrypted-file cleanup.
Last writer destroys critical changes	Version checks, typed conflicts and manual resolution for high-risk records.
Lost device exposes project files	Encrypted storage, secure keys, session/device revocation and short offline expiry for sensitive content.
Large files block all synchronization	Independent resumable file queues and command priority.
Mobile app becomes desktop UI squeezed smaller	Field-first IA, large controls, rapid capture and role-based module visibility.

3. Success Metrics and Guardrails

3.1 Product and operational metrics

Metric	Interpretation / target direction
Offline workflow completion	Share of enabled field workflows completed without requiring continuous connectivity.
Sync success rate	Accepted commands and files divided by attempted uploads, excluding business-valid rejections.
Median sync latency	Time from restored connectivity to server confirmation for standard commands.
Conflict rate	Conflicts per 1,000 commands, segmented by module and cause.
Duplicate prevention	Repeated idempotent submissions resolved without duplicate source records.
Crash-free sessions	Stability across supported OS and minimum hardware.
Pending queue age	Age distribution of unsynchronized records, with critical-item alerts.
Workspace freshness	Percentage of active workspaces synchronized within configured freshness window.
Security health	Revoked-device attempts, untrusted devices, expired workspaces and permission rejections.

3.2 Guardrails

- Offline or mobile activity metrics must not automatically become employee-performance judgments.
- GPS evidence must be purpose-limited and must not imply continuous tracking or misconduct by itself.
- No UI may display synchronized, delivered or accepted until server confirmation exists.
- Background execution must be described as best effort because mobile operating systems may suspend applications.
- Cached data volume must be minimized and remain within explicit workspace scopes.

4. Scope, Non-Goals and Assumptions

4.1 In scope

- Responsive web, PWA, native mobile and tablet architecture.
- Encrypted offline workspaces, local database, file container, manifests and local search.

- Delta synchronization, command queues, file upload queues, idempotency, dependencies and conflicts.
- Device registration, trust, session control, expiry, revocation and security checks.
- Mobile workflows for Tasks, Work Progress, QA/QC, HSE, Inventory, Assets, attendance, timesheets, Documents, drawings, BIM, Calendar, Procurement and Portals.
- Low-bandwidth mode, media capture, time/GPS evidence, QR/barcode and accessibility.
- APIs, events, audit, telemetry, testing and phased delivery.

4.2 Non-goals

- Replacing source modules or creating a separate mobile business database.
- Finalizing payroll, financial posting, approvals, contract acceptance or other high-risk decisions offline.
- Guaranteeing continuous background sync on all operating systems.
- Guaranteeing prevention of screenshots or external photography.
- Continuous employee location tracking by default.
- Full authoring-model downloads or desktop-grade BIM authoring on mobile.
- Initial dependence on external MDM, cloud storage, biometrics hardware or third-party field platforms.

4.3 Assumptions

- Every source module exposes stable mobile projections and command handlers.
- Permissions, business validation and audit remain server-side authorities.
- Native mobile is required for the strongest offline and device-security capabilities.
- PWA offline support is constrained by browser and operating-system behavior.
- Platform Configuration defines which workflows and data classes may be offline.

5. Terminology and Core Concepts

Term	Definition
Mobile projection	Permission-safe, purpose-built representation of a source record for mobile use.
Offline workspace	Controlled package of records, files and configuration available on one authorized device.
Workspace manifest	Signed inventory of workspace scope, versions, checksums, expiry and security classification.
Offline command	User action captured locally and submitted for source-module validation.
Sync session	Audited exchange of deltas, commands, files, acknowledgements and checkpoints.
Idempotency key	Unique key ensuring repeated submission produces one business result.
Conflict	A condition where local intent cannot be safely applied to current server state automatically.
Local access grant	Device-bound permission snapshot with scope and expiry; never a substitute for server authorization.
Pending upload	Encrypted attachment or chunk awaiting transfer and server validation.
Server confirmation	Definitive response that a command or file was accepted, rejected or conflicted.

6. Permanent Source-of-Truth Boundaries

Capability	Authoritative owner	Mobile & Offline responsibility
Tasks	Tasks Module	Cache assigned projections; capture commands, comments, evidence and local state.
Progress	Work Progress	Capture pending progress submissions and media; synchronize for validation.
Quality	QA/QC	Provide checklists and evidence capture; final acceptance remains server-side.
Safety	HSE	Capture hazards/incidents and prioritize sync; classification and investigation remain HSE.
Inventory	Inventory	Capture pending counts/issues/returns; stock validation remains server-side.
Assets	Assets	Capture readings, inspections and requests; lifecycle state remains Assets.
Attendance/payable time	HR & Payroll	Capture events/timesheets; payroll locking and calculation remain server-side.
Documents	Documents	Cache controlled revisions and upload drafts; document identity/versioning remain Documents.
Financial/contract decisions	Finance/Contracts	View authorized projections only; no offline finalization.
Notifications	Notifications & Event Centre	Receive minimal push payloads and retrieve details after authorization.

Boundary enforcement: The mobile database may store projections and drafts, but it may never become the system of record for an operational entity.

7. Personas and Stakeholders

Persona	Primary mobile needs
Site engineer	Tasks, progress, drawings, inspections, defects, daily evidence.
HSE officer	Hazards, incidents, inspections, toolbox talks, emergency information.
QA/QC inspector	Checklists, material/work inspections, NCR and reinspection evidence.
Storekeeper	Barcode lookup, receipt, issue, return, count and transfer drafts.
Asset operator/manager	QR lookup, readings, fuel, condition, inspection and damage reports.
Foreman/supervisor	Crew work, quantities, photos, task status, labour hours and delivery checks.
Employee	Attendance, timesheet, roster, leave and payslip viewing under policy.
Project manager	Mobile dashboards, approvals later, conflicts, workspaces and alerts.
Client user	Progress, unit, documents, defects and warranty drafts through portal projection.
Supplier user	PO, delivery, submittal, QA/HSE evidence and invoice status through portal APIs.

Persona	Primary mobile needs
Platform administrator	Device policy, version support, security, workspace limits and monitoring.

8. Product Surfaces and Device Strategy

Surface	Primary role	Offline capability
Responsive web	Universal online access and tablet administration	Application cache and limited draft support only.
PWA	Installable lightweight mobile experience	Controlled offline forms and cache subject to browser limits.
Native mobile	Primary advanced field application	Encrypted database, background sync, push, device trust and large resumable files.
Tablet	Drawings, BIM, inspections, daily reports and meetings	Same native foundation with tablet layouts and larger packages.
Client/Supplier mobile portal	External relationship-based experience	Only portal-safe projections and configured offline drafts.

8.1 Support policy

- Define minimum supported iOS, Android, browser and hardware versions through Platform Configuration.
- Maintain explicit application/API compatibility and deprecation windows.
- Block synchronization for security-critical unsupported versions.
- Use capability negotiation so older clients do not receive unsupported commands or fields.

9. Information Architecture and Navigation

Primary route	Purpose
Home	Today, attention, sync state and quick actions.
My Work	Assigned tasks, inspections, actions and due work.
Projects	Downloaded and online project workspaces.
Site	Progress, locations, daily reports and deliveries.
QA/QC	Inspections, defects, NCR and snagging.
HSE	Hazards, incidents, inspections and emergency information.
Documents	Cached revisions, drawings, uploads and required reading.
Inventory / Assets	Scanning, transactions, readings and evidence.
Timesheets	Attendance, hours and roster.
Notifications	Permission-safe alerts and acknowledgements.
Upload Queue	Pending commands/files, conflicts and failures.
Offline Workspaces	Scope, freshness, storage, expiry and removal.
Settings	Device, sync, storage, quality, notifications and security.

Navigation rule: Only modules enabled by Platform Configuration and allowed by the current user, company, project and

device context are displayed.

10. Functional Requirements Register

ID	Area	Requirement	Priority	Acceptance signal
MOB-001	Surfaces	The platform shall provide responsive web, PWA and native mobile delivery using shared domain contracts and permission rules.	Must	Supported surfaces render approved modules and enforce identical server authorization.
MOB-002	Navigation	The mobile shell shall build navigation from permissions, enabled modules, workspace scope and device capability.	Must	Unauthorized routes are absent and direct URL/deep-link access is denied.
MOB-003	Projection	Source modules shall expose mobile-safe projections rather than unrestricted internal entities.	Must	Mobile payloads contain only registered fields and pass leakage tests.
MOB-004	Workspace	Users shall download only explicitly scoped offline workspaces.	Must	Workspace manifest lists scope, records, versions, checksums and expiry.
MOB-005	Storage	All controlled local records and files shall be encrypted at rest.	Must	Security tests confirm no readable controlled payload in unencrypted storage.
MOB-006	Commands	Offline mutations shall be represented as commands with idempotency keys and expected server versions.	Must	Repeated submission creates one source result and returns deterministic response.
MOB-007	Sync	The sync engine shall support delta downloads, queued uploads, resumable files and checkpoints.	Must	Interrupted sessions resume without lost or duplicated work.
MOB-008	Priority	Critical HSE and attendance events shall receive configurable queue priority.	Must	Critical records are attempted before lower-priority files while preserving dependencies.
MOB-009	Conflict	Unsafe concurrent changes shall create explicit conflict records.	Must	Critical conflicts are visible and never resolved silently.
MOB-010	Permission	Server permissions shall be rechecked before applying every command and file.	Must	Revoked users cannot synchronize unauthorized changes.
MOB-011	Expiry	Offline workspaces and sensitive files shall support configurable expiry and removal.	Must	Expired data is blocked and removed according to policy.
MOB-012	Device	Every native installation shall register a device record and security state.	Must	Unknown/revoked devices cannot synchronize.

ID	Area	Requirement	Priority	Acceptance signal
MOB-013	Revocation	Administrators shall revoke sessions, devices and offline access.	Must	Revoked device is blocked and receives data-removal instruction on reconnection.
MOB-014	Evidence	Media capture shall retain checksum, device, local/server timing and contextual links.	Must	Evidence chain is visible in the source record and audit.
MOB-015	Time	The system shall preserve device time, time zone, last server time and estimated offset.	Must	Clock anomalies are flagged without overwriting original timestamps.
MOB-016	GPS	GPS shall be optional, purpose-limited and configured per workflow.	Must	No continuous tracking occurs unless separately authorized and designed.
MOB-017	Scanning	QR/barcode scans shall resolve stable identifiers and recheck access.	Should	Unauthorized scanned records remain inaccessible.
MOB-018	Files	Large files shall upload asynchronously in resumable chunks.	Must	Network interruption resumes from accepted chunk boundary.
MOB-019	Status	The UI shall distinguish local, pending, uploading, accepted, rejected and conflicted states.	Must	No record displays accepted before server confirmation.
MOB-020	Offline search	Local search shall index only active workspace projections and permitted metadata.	Must	Revoked/expired records disappear from search.
MOB-021	Low bandwidth	A low-bandwidth mode shall reduce media and prioritize deltas and forms.	Should	Transfer size and deferred items are visible to the user.
MOB-022	Updates	The platform shall enforce minimum application and schema versions.	Must	Blocked versions cannot sync and show clear upgrade action.
MOB-023	Audit	Device, workspace, command, file, conflict and revocation events shall be auditable.	Must	Audit provides user, device, time, action, result and correlation ID.
MOB-024	External portal	Portal mobile experiences shall use external portal APIs and projections only.	Must	Internal API routes reject portal tokens.

11. Mobile Home and Attention Centre

11.1 Required widgets

- My tasks
- Today's schedule
- Active project
- Required inspections
- Progress updates due
- Open defects
- HSE actions
- Expected deliveries

- Missing timesheet
- Notifications
- Sync status
- Offline storage status

11.2 Attention states

State	Required behavior
Unsynchronized critical record	Pinned warning with retry and local-preservation confirmation.
Failed upload	Show reason, retry eligibility and whether source record exists.
Conflict	Open comparison and allowed resolution actions.
Expired workspace	Block controlled data and request online refresh.
Storage nearly full	Show category usage and safe cleanup actions.
Permission removed	Redact/remove cached content and explain that local draft was rejected or preserved for support.
Unsupported app version	Block sync when required and direct to update.

12. Offline Workspace Model

12.1 Workspace types

- Project workspace
- Site workspace
- Building, floor or zone workspace
- Inspection workspace
- Delivery workspace
- Task package
- HSE package
- Handover package
- Client defect package
- Supplier submission package

12.2 Workspace content controls

Content class	Examples	Policy controls
Structured projections	Tasks, locations, work packages, assets, inventory references	Scope, field list, date range, status and max record count.
Controlled files	Drawings, procedures, forms, selected documents	Classification, revision, watermark, expiry and download mode.
Templates	Checklists, forms, submission schemas	Template version, effective date and migration behavior.
Model fragments	Floor/zone BIM geometry and metadata	LOD, texture quality, element permissions and storage limit.
Search index	Permitted names, codes and metadata	Encrypted, scope-limited and removed on expiry/revocation.

13. Workspace Lifecycle, Manifest and Packaging

Requested

```

-> Permission Checked
-> Preparing Package
-> Downloading
-> Available Offline
-> Updating
-> Expiring
-> Expired / Removed

```

13.1 Signed manifest

```

workspace_id
manifest_version
tenant_id
user_id
device_id
scope
included_entities
record_versions
file_checksums
security_classification
generated_at
expires_at
encryption_key_reference
signature

```

13.2 Packaging requirements

- Generate package only after current user, device, tenant, company, project and record permission checks.
- Use encrypted transfer and encrypted local storage keys bound to device trust.
- Support resumable package download and incomplete-package recovery.
- Validate signature and checksums before making content available.
- Discard partial content securely when validation fails.

14. Selective Synchronization and Download Scope

Scope dimension	Examples
Assignment	Assigned tasks, inspections, assets or responsible work packages.
Project structure	Project, building, floor, zone, unit or location.
Time	Today, current week, active reporting period or configured history.
Operational domain	QA/QC only, HSE only, deliveries only or mixed field package.
Files	Metadata only, selected folders, current drawings or full approved package.
Connectivity policy	Automatic, manual, Wi-Fi only, low-bandwidth or emergency.

Data minimization: The application must never download an entire tenant or company dataset merely because the user belongs to that company.

15. Local Data and Storage Architecture

15.1 Local components

- Encrypted relational database
- Encrypted attachment container
- OS secure credential/key store
- Sync queue database
- Encrypted local search index
- Temporary media cache
- Configuration and feature-capability cache

15.2 Offline record fields

```

local_record_id
server_record_id
tenant_id
company_id
project_id
entity_type
server_version
local_version
sync_state
permission_snapshot
downloaded_at
last_modified_at
created_offline
created_by_device
security_classification
expires_at

```

15.3 Storage classes

Class	Behavior
Persistent authorized projection	Retained until expiry, removal or revocation.
Pending user draft	Retained until synchronized, explicitly discarded or administratively resolved.
Pending upload	Protected from automated cleanup until server confirmation.
Temporary cache	Safe to remove automatically under configured policy.
Restricted no-backup	Excluded from OS/cloud backup and external file browsing.
Recoverable synchronized content	May be removed locally and downloaded again.

16. Offline Command Model

Offline commands represent user intent rather than copied database state. Each command is validated by the owning source module against the latest server record, permissions, status, template and business rules.

```

command_id
command_type
user_id
device_id
entity_type
entity_id_or_local_reference
payload
local_timestamp
expected_server_version
dependencies
evidence_references
idempotency_key
status

```

16.1 Example commands

- CreateProgressUpdate
- CompleteTask
- CreateInspection
- SubmitChecklist
- CreateDefect
- CreateHSEObservation
- RecordStockIssue
- RecordAssetReading
- ClockIn
- SubmitTimesheet
- UploadDocumentRevision
- CreateDeliveryReceiptDraft

16.2 Command statuses

Status	Meaning
Draft	Not queued for synchronization.
Pending	Ready to send when policy permits.
Sending	Included in active sync session.
Accepted	Applied by owning source module.
Rejected	Business or permission validation failed.
Conflict	User/administrator resolution required.
Cancelled	Explicitly withdrawn before acceptance.
Expired	No longer valid under policy.

17. Synchronization Sessions and Queue

17.1 Download flow

```
Device requests delta
-> authenticate user and device
-> revalidate scope and permissions
-> identify changed projections/files
-> generate signed delta
-> download and verify
-> update local database and checkpoint
```

17.2 Upload flow

```
Pending command/file
-> connectivity and policy check
-> dependency ordering
-> permission revalidation
-> expected-version validation
-> source-module business validation
-> accepted / rejected / conflict
-> return server record and audit result
```

17.3 Queue priorities

1. Emergency and critical HSE records
2. Attendance events
3. Inspection and quality records
4. Progress updates
5. Task updates
6. Inventory and asset transactions
7. Documents and media
8. Non-critical analytical data

Priority rule: Priority changes transmission order only. It does not bypass permissions, dependencies, business validation or source-module ownership.

18. Idempotency, Dependencies and Ordering

18.1 Idempotency requirements

- Every command and final file-completion request carries a client-generated unique idempotency key.
- The server stores the processing result for a defined retention period.
- Repeated requests return the original accepted/rejected result without repeating business effects.
- Idempotency scope includes tenant, command type and initiating identity.

18.2 Dependency graph

```
CreateInspection
-> UploadPhoto A
-> UploadPhoto B
-> SubmitInspection
-> CreateDefect
```

- Dependencies use local references until server IDs are returned.
- Child actions wait for parent acceptance.
- Parent rejection blocks dependants and shows a recoverable state.
- Unrelated commands must continue even when one dependency chain fails.

19. Conflict Detection and Resolution

19.1 Conflict classes

Conflict	Example	Default handling
Version conflict	Task changed on server after offline edit	Manual or safe field-level merge.
Status conflict	Inspection closed while offline response pending	Server wins; preserve local evidence as draft.
Permission conflict	User removed from project	Reject and revoke cached access.
Template conflict	Checklist revised while old template cached	Prompt migration or resubmission on current template.
Structure conflict	Location or work package archived	Require new target or discard.
Duplicate business record	Same defect created online and offline	Suggest link/merge; never auto-delete evidence.
Document revision conflict	Cached revision superseded	Warn and require upload against current document rules.

19.2 Resolution strategies

- Server wins for permissions, locks, payroll, financial posting, approved records and published documents.
- Preserve client draft when user content/evidence remains useful.
- Automatic merge only for explicitly registered commutative or independent fields.
- Manual resolution where business meaning, liability, quantity or acceptance may change.

19.3 Conflict UI

- Local and server values
- Changed fields
- Actors and timestamps
- Rejection/conflict reason
- Allowed actions
- Audit/support reference

20. Permissions, Expiry and Revocation

20.1 Required checks

9. Before workspace preparation
10. Before opening high-sensitivity cached content where online revalidation is required
11. Before every command is applied
12. Before every attachment is linked
13. Before every notification deep link opens

20.2 Revocation behavior

- Mark local access grant revoked and prevent record opening.
- Remove or cryptographically invalidate restricted files and search index entries.
- Reject unauthorized pending commands while preserving necessary audit/support metadata.
- Invalidate cached previews and thumbnails.
- Keep source audit records; do not erase evidence of prior authorized access.

20.3 Expiry policy examples

Data class	Example policy
Standard project projections	Configurable days since last successful online validation.
Payroll or executive compensation	Offline disabled or very short expiry.
Legal/contract restricted documents	Explicit download permission and short effective window.
Emergency procedures	Longer controlled availability with revision/freshness checks.
RFQ package	Expires at tender closure or relationship revocation.
Client/supplier projections	Expires with the active portal relationship.

21. Device Registration, Trust and Lifecycle

Unregistered -> Registration Pending -> Trusted -> Active -> Restricted -> Revoked -> Archived

21.1 Device record

- Device ID
- User
- Platform and OS version

- Application version
- Device model
- Registration and last-active dates
- Push token
- Encryption and biometric capabilities
- Root/jailbreak status
- Trust status
- Remote-revocation status

21.2 Trust rules

- Unknown devices cannot receive offline workspaces.
- Device registration does not replace user authentication or MFA.
- High-risk projects may require explicit administrator approval for offline access.
- Application attestation and integrity signals may restrict but should not silently delete user drafts.

22. Device Security and Lost-Device Response

22.1 Required controls

- Encrypted local database and file container
- OS keychain/secure enclave
- Application PIN and optional biometric unlock
- Automatic lock
- Session and device revocation
- Root/jailbreak detection
- Debugging restrictions
- Certificate pinning where appropriate
- No controlled payload in logs
- Clipboard/screenshot controls where supported and configured

22.2 Lost or stolen device actions

Action	Expected result
Revoke sessions	Prevent new API and sync calls.
Revoke device	Invalidate local access keys and push token.
Mark lost	Raise security state and block workspace renewal.
Remote removal instruction	Delete/cryptographically invalidate controlled data on next connection.
Force credential reset	Protect account where compromise is suspected.
Review activity	Expose recent sessions, downloads, failed access and last sync.

Security limitation: The platform cannot guarantee recovery of files deliberately copied outside its controlled container. Offline access must therefore be minimized and time-limited.

23. Media Capture and File Processing

23.1 Supported capture

- Photo
- Video
- Audio note where enabled
- Document scan

- Signature
- QR/barcode
- GPS/location
- Timestamp
- Drawing/photo annotation

23.2 Processing pipeline

```

Capture
-> local validation
-> compression profile
-> checksum
-> metadata attachment
-> encryption
-> upload queue
-> resumable transfer
-> server file/malware validation
-> source-record link

```

23.3 Evidence metadata

Field group	Required values
Identity	Captured by, device, tenant, company and project.
Time	Device time, time zone, last server time, adjusted estimate and upload time.
Location	Coordinates, accuracy, timestamp and purpose where authorized.
Integrity	Checksum, file size, original/annotated indicator and transformations.
Context	Entity, location, checklist item, work package or submission reference.

24. Time, GPS, Geofence and Evidence

24.1 Time integrity

- Store original device time and never overwrite it.
- Store last known server time and calculated device offset.
- Flag significant clock drift.
- Use server time for final processing/receipt while retaining local capture time.

24.2 GPS and geofence controls

Control	Requirement
Purpose limitation	Location is requested only for configured workflows.
Consent and notice	User sees why location is required and retention policy.
Accuracy	Store accuracy and distinguish unavailable/low-confidence fixes.
Continuous tracking	Disabled by default; separate design and legal review required.
Geofence	Use as evidence/warning, not automatic proof of performance or misconduct.
Fallback	Provide documented exception process when GPS is unavailable.

25. QR, Barcode and Local Search

25.1 Scanning use cases

- Asset lookup
- Inventory item and bin lookup
- Delivery lookup
- Document lookup
- Unit/location lookup
- Inspection location
- Configured employee/site badge

25.2 Security rules

- Codes contain stable IDs or secure URLs, not confidential payloads.
- Scan resolution always performs local-scope and server authorization checks.
- Unauthorized records never reveal titles, counts or metadata.
- Offline scan may open only records already present in an active workspace.

25.3 Local search

- Indexes only active workspace fields.
- Uses encrypted storage.
- Supports project, task, location, drawing, asset, inventory, document and checklist lookup.
- Removes expired/revoked entries and excludes hidden metadata.

26. Tasks Mobile Requirements

ID	Area	Requirement	Priority	Acceptance signal
TSK-M01	Tasks	Users shall view assigned task projections offline.	Must	Assigned tasks open in active workspace with current cached version.
TSK-M02	Tasks	Users shall change permitted task states through offline commands.	Must	Server validates workflow state and version before applying.
TSK-M03	Tasks	Users shall add comments, checklist responses, time and evidence offline.	Must	Each item synchronizes independently and retains source context.
TSK-M04	Tasks	Closed, deleted or reassigned tasks shall not be silently reopened.	Must	Conflict/rejection is shown and local content can be preserved as draft.
TSK-M05	Tasks	Approval decisions shall not be finalized offline.	Must	Offline UI offers draft/comment only for approval-bound actions.

27. Work Progress Mobile Requirements

- Select project structure and work package
- Enter completed quantity and unit
- Record crew/labour, materials and equipment
- Attach photos and notes
- Prepare daily progress submission
- Display pending vs accepted progress clearly

Validation	Behavior
Quantity limits	Validate locally from cached rules and revalidate on server.
Duplicate reporting	Detect local duplicates and server overlap.
Location/work-package status	Reject archived/closed targets or create conflict.
Accepted-progress boundary	Offline submission cannot directly change certified/accepted totals.
Evidence	Preserve local and server receipt timestamps and file checksums.

28. QA/QC Mobile Requirements

- Inspection request and checklist
- Material/work inspection
- Defect and NCR evidence
- Snagging
- Reinspection evidence
- Signature where permitted
- Drawing markup
- Cached BIM element selection

Quality rule: Offline completion records findings and evidence. Final acceptance, rejection, closure and quality status require source-module authorization and server validation.

28.1 Template version behavior

- Every checklist stores template version.
- Changed templates create migration or resubmission flow.
- Submitted inspection revisions remain immutable.
- Required questions cannot disappear silently during synchronization.

29. HSE Mobile Requirements

- Hazard observation
- Near miss
- Initial incident report
- Safety inspection
- Toolbox talk attendance
- Corrective-action evidence
- Stop-work request
- Cached emergency procedures and contacts

29.1 Emergency safeguards

- Critical HSE command receives top queue priority.
- The UI confirms local recording separately from server receipt.
- Emergency calling remains available without sync.
- Local evidence remains until server acknowledgment.
- Failure/retry state remains pinned and visible.

No false delivery: The application must never imply that management received an offline emergency report before the server confirms receipt.

30. Inventory Mobile Requirements

- Item and bin lookup
- Stock count
- Issue draft
- Return draft
- Transfer draft
- Delivery receipt draft
- Barcode scan
- Photo evidence

Concurrency case	Required response
Stock consumed by another user	Reject or adjust through explicit conflict; do not create negative stock silently.
Closed financial/stock period	Reject command and preserve user evidence.
Reservation changed	Revalidate against current reservation and available stock.
Duplicate scan/receipt	Use idempotency and business duplicate detection.

31. Assets Mobile Requirements

- QR lookup
- Condition check
- Meter reading
- Fuel record
- Maintenance evidence
- Inspection checklist
- Damage report
- Transfer request
- Assignment acknowledgement

Asset rule: Offline data creates readings, evidence or requests. Asset ownership, custody, availability, maintenance state and transfer remain authoritative in Assets.

32. Attendance and Timesheets Mobile Requirements

32.1 Attendance events

- Clock in/out
- Break start/end
- Project/site selection
- Optional GPS evidence
- Optional QR/site validation
- Manual exception explanation

32.2 Required statuses

Displayed state	Meaning
Recorded locally	Saved on device; not delivered.
Awaiting synchronization	Queued and protected locally.
Accepted by server	Authoritative attendance event exists.

Displayed state	Meaning
Rejected/conflict	Requires explanation or administrator review.

32.3 Timesheets

- Daily hours by project/work package/role
- Overtime and notes
- Supporting evidence
- Local total validation
- No offline payroll lock or formal approval

33. Documents and Drawings Mobile Requirements

33.1 Documents

- View cached authorized revision
- Download selected document
- Search safe metadata
- Annotate local copy
- Upload new file/revision draft
- Create document request
- Acknowledge required reading

33.2 Drawing viewer

- Optimized PDF/DWG-derived output
- Zoom/pan
- Layers where supported
- Pins and markups
- Measurements
- Revision comparison
- Issue/photo linking
- Offline drawing packages

33.3 Revision safeguards

- Display cached/offline indicator and revision number.
- Warn when a newer revision is known.
- Block use of withdrawn documents where policy requires.
- Preserve annotation against the original revision and support controlled migration.

34. BIM/3D Mobile Requirements

- Stream or download optimized model fragments
- Select and isolate element
- View permission-safe metadata
- Filter floor/zone
- Open linked tasks/defects
- Create field issue
- View progress state
- Cache selected zones offline

Constraint	Requirement
Geometry	Optimized, compressed and level-of-detail based.
Metadata	Only fields permitted for the user and workflow.
Textures	Configurable quality and low-bandwidth reduction.

Constraint	Requirement
Scope	Selected project zones; no full model by default.
Revision	Package identifies model version and obsolete-package warning.
Storage	Per-device/project quota and explicit removal.

35. Calendar, Meetings and Notifications

35.1 Calendar and meetings

- View cached schedule and assigned meetings
- Accept/decline online or queue controlled response
- View agenda and attached cached documents
- Capture personal notes and field evidence
- View issued minutes and acknowledge decisions

35.2 Notifications

- Receive minimal push payload
- Authenticate before retrieving details
- Maintain in-app feed and badge count
- Support deep links into mobile routes
- Cache safe notification summary offline
- Support emergency alerts and acknowledgements under Notifications policy

36. Procurement and Delivery Workflows

- Expected delivery list
- PO/delivery lookup
- Arrival and dispatch times
- Received quantity and condition
- Packing list/certificates
- Photos
- Rejection evidence
- Supplier and receiver signatures where configured

Receiving rule: Mobile capture creates a pending receipt. Procurement and Inventory validate the final delivery and stock movement after synchronization.

37. Client and Supplier Portal Mobile Workflows

Portal	Mobile priorities	Offline boundary
Client	Progress, unit, payments, documents, defects, warranty, inspection scheduling	Draft defect/warranty/evidence only; portal projection remains source view.
Supplier	RFQ alert, PO, delivery, submittal, timesheet, QA/HSE response, invoice status	Configured submission drafts; internal APIs and records remain inaccessible.

Token boundary: Portal tokens may call only portal/mobile projection APIs. Internal mobile APIs must reject external identities.

38. Low-Bandwidth and Background Behavior

38.1 Low-bandwidth mode

- Load text and forms before media
- Use thumbnails
- Delay large files
- Compress images
- Disable automatic video download
- Use delta sync
- Reduce analytics payloads
- Pause BIM textures
- Show estimated transfer size

38.2 Background sync conditions

- Valid session
- Trusted device
- OS permits background work
- Network and Wi-Fi policy permits
- Battery/data policy permits
- Workspace not expired

Operating-system limitation: Background synchronization is best effort. The product must surface the last actual successful sync and pending queue rather than promise continuous execution.

39. Application Updates and API Compatibility

Update policy	Behavior
Optional	User may update; current client remains supported.
Recommended	Persistent banner and release notes; sync remains available.
Required	Block selected features or synchronization after deadline.
Security-blocking	Immediate sync block with clear update route.
Schema migration	Backup/transactional migration and rollback-safe startup.

39.1 Compatibility requirements

- Versioned mobile APIs
- Capability negotiation
- Explicit deprecation window
- Backward-compatible event schemas where supported
- Workspace manifest version
- Local database schema migration
- Minimum OS and application versions

40. Data Model

RegisteredDevice
DeviceSession
DeviceSecurityState
OfflineWorkspace

```

WorkspaceScope
WorkspaceManifest
WorkspaceRecord
WorkspaceFile
LocalAccessGrant
SyncSession
SyncCheckpoint
SyncDelta
OfflineCommand
OfflineCommandDependency
CommandProcessingResult
SyncConflict
ConflictResolution
PendingUpload
UploadChunk
DeviceRevocation
MobileConfiguration
MobileFeatureCapability
MobileTelemetryEvent
MobileAuditEvent

```

40.1 Entity requirements

- Tenant and company scope on all entities
- Device and user binding where relevant
- Version, created/updated timestamps and correlation IDs
- Retention and expiry fields
- No hard deletion of auditable command results
- Encryption-key references without storing raw keys in business tables

41. Backend Services and Domain Events

41.1 Recommended services

Service	Responsibility
Mobile API Gateway	Authentication, device context, rate limits, version and routing.
Device Management	Registration, trust, security state and revocation.
Offline Workspace	Scope validation, manifest, package and expiry.
Mobile Projection	Generate permission-safe records from source modules.
Sync Orchestration	Sessions, checkpoints, deltas, queue and result aggregation.
Offline Command	Validate envelope and route to source-module handlers.
Conflict Resolution	Classify, expose and apply approved resolutions.
Mobile File / Resumable Upload	Chunk transfer, checksums, finalization and source linking.
Push Adapter	Device token management and minimal notification payloads.
Telemetry / Audit	Operational metrics, crashes, security and full audit events.

41.2 Event examples

```

Consumed:
UserPermissionChanged
ProjectAssignmentChanged
TaskAssigned
DocumentRevisionPublished
WorkspaceDataChanged

```


DeviceRevoked
ApplicationVersionBlocked

Published:
MobileDeviceRegistered
OfflineWorkspaceCreated
OfflineCommandReceived
OfflineCommandAccepted
OfflineCommandRejected
MobileConflictCreated
MobileFileUploaded
OfflineAccessRevoked
DeviceSecurityFailed

- Use transactional outbox
- Idempotent consumers
- Schema versioning
- Correlation and causation IDs
- Retry queues
- Dead-letter handling

42. API Contract

42.1 Device management

POST /mobile/v1/devices/register
GET /mobile/v1/devices/current
POST /mobile/v1/devices/current/verify
POST /mobile/v1/devices/current/revoke

42.2 Workspaces

GET /mobile/v1/workspaces
POST /mobile/v1/workspaces
GET /mobile/v1/workspaces/{id}/manifest
GET /mobile/v1/workspaces/{id}/delta
DELETE /mobile/v1/workspaces/{id}

42.3 Synchronization and conflicts

POST /mobile/v1/sync/sessions
POST /mobile/v1/sync/commands
GET /mobile/v1/sync/commands/{id}
GET /mobile/v1/sync/conflicts
POST /mobile/v1/sync/conflicts/{id}/resolve

42.4 Files

POST /mobile/v1/uploads
PUT /mobile/v1/uploads/{id}/chunks/{chunk}
POST /mobile/v1/uploads/{id}/complete
GET /mobile/v1/files/{id}

42.5 Write-contract requirements

- Idempotency key
- Device identity
- User session

- Local timestamp and time zone
- Expected server version
- Permission context
- Correlation ID
- Audit metadata
- Schema/capability version

43. Frontend and Native Application Architecture

43.1 Recommended client layers

Layer	Components
Application shell	Navigation, authentication, project switcher, online/offline state.
Feature modules	Tasks, QA/QC, HSE, progress, documents, inventory, assets and timesheets.
Local data layer	Encrypted repository, projection stores and local search.
Command layer	Command creation, validation, dependencies and optimistic local state.
Sync layer	Sessions, queue, retry, delta apply, upload and conflict.
Device layer	Camera, files, GPS, QR, notifications, biometrics and secure storage.
Telemetry layer	Crash, performance, sync, security and support references.

43.2 State principles

- Server state and local draft state remain distinguishable.
- Optimistic updates are permitted only when the UI clearly marks them pending.
- Critical acceptance/approval states are never optimistic.
- Each list item exposes sync state without opening details.
- One failed feature or queue item must not break unrelated workflows.

44. UX States, Error Handling and Accessibility

44.1 Standard UX states

- Online current
- Offline cached
- Offline draft
- Pending sync
- Uploading
- Accepted
- Rejected
- Conflict
- Expired
- Permission revoked
- Unsupported version
- Storage blocked

44.2 Error categories

Category	User message must explain
Connectivity	Local data preservation and automatic/manual retry.

Category	User message must explain
Authentication	Whether offline viewing remains available and how to reauthenticate.
Permission	Scope removed and disposition of local draft.
Validation	Specific business field/rule and correction action.
Conflict	Local/server difference and permitted resolutions.
Storage	Required space and safe cleanup options.
File	Unsupported type, size, checksum or malware failure.
Device security	Required action and whether sync is blocked.
Server unavailable	Last successful sync and queue preservation.

44.3 Accessibility

- Screen-reader labels
- Large touch targets
- Text scaling
- High contrast
- No color-only status
- Reduced motion
- Accessible form errors
- Landscape/portrait support
- Keyboard support on tablets
- Controls usable with gloves where practical

45. Security, Privacy, Retention and Audit

45.1 Security requirements

- Tenant/company/project isolation
- Encrypted local and transport data
- Field-level projection filtering
- Secure device keys
- Session/device revocation
- Minimal push content
- No confidential logs
- Rate limits and abuse protection
- Malware/file scanning
- Root/jailbreak policy
- Short offline expiry for sensitive data

45.2 Privacy requirements

- Data minimization
- Purpose-limited GPS
- Configurable retention
- No continuous tracking by default
- Consent/notice where required
- Controlled deletion or cryptographic invalidation
- Legal hold compatibility for synchronized authoritative records

45.3 Audit events

- Device registered/trusted/revoked
- Workspace created/downloaded/removed
- Command created/sent/accepted/rejected
- Conflict created/resolved

- File captured/uploaded/rejected
- Permission revoked
- Security check failed
- Emergency submission attempted
- Application version blocked

46. Administration and Platform Configuration

Configuration domain	Options
Surface and modules	Enabled platforms, modules, routes and widgets.
Device support	Minimum OS/app versions, trust, integrity and biometric rules.
Offline data	Enabled entity types, scopes, record limits, expiry and freshness.
Storage and files	Device quota, image/video quality, file size, Wi-Fi-only rules.
Synchronization	Frequency, background behavior, retries, priorities and delta limits.
Conflicts	Automatic merge allow-list, server-wins classes and escalation roles.
Location	GPS requirements, geofences, accuracy, retention and exceptions.
Security	Session/offline duration, screenshots, clipboard and revocation behavior.
UX	Navigation, quick actions, default project, terminology and custom fields.
Notifications	Push types, emergency behavior, badges and acknowledgement hooks.

Configuration boundary: Configuration may enable or restrict a capability, but it never overrides the owning source module's business validation or the user's permissions.

47. Observability and Analytics

47.1 Technical metrics

- Active/registered devices
- Application and OS versions
- Sync success and latency
- Command rejection/conflict rate
- Upload failure and resume rate
- Queue age
- Workspace size/freshness
- Crash-free sessions
- Background sync attempts
- Revoked-device access attempts

47.2 Operational metrics

- Progress submissions
- Inspections
- HSE reports
- Attendance events
- Delivery receipts
- Inventory transactions
- Asset readings

- Offline usage by project

Analytics guardrail: These metrics support product and operational health. They must not automatically become employee performance or disciplinary evidence.

48. Non-Functional Requirements

Category	Requirement
Performance	App shell under 2 seconds and cached common form/list under 1 second on defined minimum hardware.
Sync	Standard delta under 10 seconds under normal connectivity; large files asynchronous and resumable.
Resilience	Pending commands survive app restart, device reboot and transient server/network failure.
Data integrity	No duplicate business effects; no synchronized label before server confirmation.
Scalability	Support many devices, projects, workspaces, photos and concurrent sync sessions.
Availability	Cached permitted workflows continue during platform outage.
Security	Encrypted storage, device trust, expiry, revocation and permission revalidation.
Accessibility	WCAG-aligned mobile interaction, text scaling, screen reader and reduced motion.
Maintainability	Versioned APIs/schemas, modular adapters, automated migrations and observability.

49. Testing Strategy and Critical Scenarios

49.1 Required test categories

- Online-to-offline and offline-to-online transitions
- Duplicate command and retry
- Network interruption
- Device restart and app crash
- Upload resume
- Permission revocation
- Workspace expiry
- App upgrade/database migration
- Server-version conflict
- Storage exhaustion
- Clock manipulation
- GPS unavailable
- Root/jailbreak
- Lost-device revocation
- Multi-company/project/portal isolation
- Malware rejection
- Low bandwidth
- Background sync
- Accessibility
- Battery and performance

49.2 Critical scenarios

Scenario	Expected result
HSE incident without internet	Local record and evidence preserved; clear not-delivered warning; priority sync and server receipt confirmation.
Power loss during upload	Accepted chunks retained; resume without duplicate file or lost command.
Task cancelled on server	Offline completion rejected/conflicted; local notes/evidence preserved.
Inspection template changed	Template version conflict with migration/resubmission path.
Project access removed	Cached access revoked and pending unauthorized commands rejected.
Same command uploaded three times	One business effect and deterministic repeated response.
Two devices edit same record	Version conflict; configured merge or manual resolution.
Device clock wrong by six hours	Original time retained; offset warning and server receipt time stored.
Document superseded offline	Clear stale revision warning and controlled annotation migration.
Storage fills during video capture	Safe failure, partial cleanup and no loss of unrelated drafts.
Device reported stolen	Sessions/device revoked; sync blocked; local keys invalidated on next contact.
Inventory consumed concurrently	Stock issue conflict; no silent negative stock.
BIM package obsolete	Model-version warning and required workspace refresh.

50. Acceptance Criteria

ID	Acceptance criterion
AC-001	A user can download an authorized workspace, disconnect, complete enabled field workflows and later synchronize them without duplicate source records.
AC-002	All controlled local records and files are encrypted and inaccessible after device/workspace revocation according to policy.
AC-003	Every offline mutation is represented by a command with idempotency, expected version, device, user, time and audit metadata.
AC-004	The application clearly differentiates local, pending, accepted, rejected and conflicted states.
AC-005	No critical conflict is silently overwritten; the user sees local/server differences and permitted resolution paths.
AC-006	Permissions are revalidated at workspace generation, command application, file linking and deep-link opening.
AC-007	Emergency HSE capture works offline and never falsely claims server delivery.
AC-008	Large file uploads resume after interruption and do not block unrelated commands.
AC-009	Expired or revoked data is removed from local views, files and search indexes.
AC-010	Portal mobile sessions cannot access internal mobile APIs or data projections.

ID	Acceptance criterion
AC-011	Application/API/schema version policies block unsupported clients safely without corrupting local drafts.
AC-012	Audit and observability can trace device, workspace, command, file, conflict, retry and revocation activity by correlation ID.
AC-013	Core mobile workflows pass accessibility, minimum-device, low-bandwidth and offline resilience tests.
AC-014	The platform operates without mandatory third-party mobile, storage, MDM or field-service integrations.

51. Delivery Phases and Rollout

Phase	Scope	Exit gate
Phase 1 - Responsive foundation	Mobile shell, responsive modules, camera/file upload, notifications, device registration, online Tasks/QA/HSE/Progress.	Supported-device UX, security and online field flows accepted.
Phase 2 - Core offline engine	Encrypted database, workspaces, deltas, commands, idempotency, queues, conflicts, revocation.	End-to-end offline/online resilience and security tests pass.
Phase 3 - Field operations	Offline Tasks, Progress, QA/QC, HSE, attendance, timesheets, Inventory, Assets, Deliveries, Documents.	Module acceptance criteria and concurrency scenarios pass.
Phase 4 - Advanced mobile	Offline BIM zones, portal drafts, geofencing, QR/barcode, signatures, advanced media and analytics.	Performance/storage/device matrix and portal isolation tests pass.

51.1 Rollout strategy

14. Internal pilot on one project and limited device set
15. Expand by workflow, not by enabling every module simultaneously
16. Train field champions and support team
17. Measure queue age, conflicts, crashes and security events
18. Enable additional projects/companies through configuration
19. Retire paper/manual processes only after stable acceptance

52. Page and Flow Registry

Page / flow	Primary contents
Mobile Home	Today widgets, attention, quick actions and sync state.
My Work	Tasks, inspections, actions and filters.
Offline Workspaces	Scope, storage, freshness, expiry, update and removal.
Workspace Detail	Manifest summary, modules, files, sync history and security.
Upload Queue	Commands/files, progress, priority, retry and failures.
Conflict Centre	Local/server comparison and resolution.
Device & Security	Trust, sessions, biometric, storage and revocation status.
Field Capture	Camera/video/scan, metadata, annotation and queue.
Task Detail	Cached task, checklist, comments, evidence and pending status.

Page / flow	Primary contents
Progress Form	Location, work package, quantity, labour, equipment, materials and media.
Inspection Form	Versioned checklist, observations, signature and evidence.
HSE Report	Hazard/incident fields, emergency state and delivery confirmation.
Inventory Scan	Item/bin, quantity, transaction draft and conflict state.
Asset Scan	Asset identity, reading, condition, inspection and evidence.
Attendance	Clock actions, location/QR evidence and sync state.
Documents/Drawings	Cached revision, markup, stale warning and upload.
BIM Viewer	Optimized zone, elements, metadata, issues and package revision.
Settings	Sync, data, quality, language, notifications and security controls.

53. Risks, Open Decisions and Definition of Done

53.1 Principal risks

Risk	Mitigation
Offline scope grows too large	Workspace limits, selective sync, quotas and project packages.
Source modules implement inconsistent command semantics	Shared command envelope, adapter standards and contract tests.
Conflict burden overwhelms field users	Safe merge allow-list, clear UX and supervisor escalation.
OS background limits create stale data	Visible freshness, foreground sync and no false promises.
Sensitive data retained too long	Classification-based expiry, device keys and revocation.
BIM/media exhaust device storage	LOD, compression, quotas, Wi-Fi policy and cleanup tools.
PWA and native capability diverge	Published capability matrix and progressive enhancement.

53.2 Open product decisions

- Native framework and supported OS baseline.
- Maximum workspace size and default expiry by security class.
- Device-attestation and rooted-device enforcement policy.
- Allowed automatic-merge fields by module.
- Minimum supported hardware for BIM and large document packages.
- Digital signature scope before final Approvals & Workflow Engine.
- Country-specific privacy constraints for GPS and attendance evidence.

53.3 Definition of done

- Functional and acceptance criteria are implemented and traceable to test evidence.
- Source-module command contracts and mobile projections are versioned and documented.
- Security, privacy, threat-model and penetration-test findings are resolved or accepted by authority.
- Offline resilience, duplicate, conflict, revocation, migration and storage scenarios pass on the supported device matrix.
- Monitoring, alerting, support playbooks, release rollback and device/app-version dashboards are operational.
- No critical workflow depends on an unimplemented third-party integration.

Final product position: The Mobile & Offline Platform is a purpose-built field execution environment. It securely delivers authorized projections to devices and turns offline activity into validated, auditable commands. It is never a smaller desktop copy and never a parallel source of truth.